Pair Trading in Leveraged ETFs: Analysis of a Cross-Jurisdictional Strategy Profitability

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Abstract

This paper investigates the empirical profitability of a cross-border pairs trading strategy utilizing leveraged Exchange-Traded Funds (ETFs) listed in the US and UK that track identical underlying indices and specifically focuses on the pair of TQQQ and LQQ3. Despite the theoretical Law of One Price, we identify persistent short-term price divergences driven by jurisdictional liquidity shifts and time-zone imbalances. Utilizing a Z-score-based mean-reversion model, we demonstrate that these discrepancies offer significant market-neutral alpha, performing consistently in both bullish and bearish regimes. However, we also conduct an analysis of the impacts of transaction costs and bid-ask spreads, which we find can greatly erode profits for retail investors who do not have access to an institutional grade set up with advantage of scale.

1. Introduction

Given the majority of hedge fund activity and academic research focuses on intra-market equity pairs within the US, what are the returns from following a cross-border strategy that pairs identical indices across different jurisdictions? Is there a persistent pricing dislocation between the US and the UK for the same underlying asset, and if so, can it be explained by the structural differences between these two major markets? In this paper, we attempt to address these questions by examining the returns and risk of a specific type of statistical arbitrage known as pairs trading, utilizing a dual-listed sample of leveraged ETFs tracking the Nasdaq-100 index: TQQQ (listed in the US) and LQQ3 (listed in the UK).

During the first quarter of 2026, global equity markets experienced significant volatility and shifting liquidity regimes. Traditional long-only strategies often faced considerable drawdowns during this period. According to broader market benchmarks, the early months of 2026 were characterized by rapid price swings that challenged conventional portfolio management. Specific Equity Market Neutral (EMN) strategies, however, are designed to remain indifferent to such directional volatility. This led to a greater focus on market-neutral strategies that could provide steady returns. Pairs trading, in particular, has historically been used to navigate these conditions by isolating relative value rather than betting on market direction.

Pairs trading is typically classified as an EMN investment strategy based on short-term price reversal. Characterized in industry as a form of statistical arbitrage, the strategy involves finding two securities that follow a similar trading pattern. When the price relationship—expressed as a ratio—deviates from a specified statistical rule, positions are taken long in the relatively undervalued security and short in the overvalued security, with the expectation that the ratio will converge to its historical mean. While traditional pairs trading derives returns from the cross-autocorrelation of different stocks, our strategy focuses on the lead-lag relationship between identical indices traded in different time zones, specifically during the high-liquidity window of the London and New York overlap.

A growing body of research is focused on the performance of pairs trading in foreign and specialized markets (Gatev, Goetzmann, and Rouwenhorst 2006; Perlin 2009; Do and Faff 2010). Gatev, Goetzmann, and Rouwenhorst (2006) examined pairs trading in the US equity market from 1962 to 2002, demonstrating that the strategy generated an average annual return of up to 11% with low levels of risk and limited exposure to systematic equity factors. Do and Faff (2010) extended this study, noting that while returns remained significant, they showed a decline as market efficiency increased. Several studies have proposed modifications to the standard methodology, such as using Gaussian Markov chain models for the spread (Elliott, Van Der Hoek, and Malcolm 2005) or utilizing cointegration for pair selection (Vidyamurthy 2004). More recently, research has begun to explicitly model the impact of transaction costs on net profitability (Do and Faff 2012).

In this paper, we make four key incremental contributions to the literature on pairs trading. we provide extensive evidence on the profitability of a cross-border, identical-index strategy, pairing TQQQ and LQQ3 to eliminate the idiosyncratic company risk common in traditional equity pairs. Second, we identify a specific "time-window alpha," focusing on the 16:00 UTC (11:00 AM NY) period when both the London Stock Exchange and the Nasdaq are active, creating a unique environment for price discovery. Third, we provide a comprehensive analysis of the risks in this strategy using a multi-factor framework, specifically calculating Greeks such as Beta and Alpha to measure market neutrality. Finally, we provide an analysis

of the strategy’s sensitivity to market frictions, incorporating price impact and bid-ask spread estimates to evaluate the robustness of the returns.

In this study, we examine the performance of the TQQQ/LQQ3 pair from April 2024 to March 2026. Over this sample period, our results demonstrate that the strategy delivered a cumulative total return of approximately 136%. Notably, the average annual standard deviation for the pairs trading portfolio was significantly lower than that of the underlying Nasdaq-100 index. These results suggest that by utilizing a Z-score-based mean-reversion model, the strategy was able to capture significant absolute returns during a period characterized by both falling equity prices and spikes in volatility. Unlike previous US studies that report a drop in performance through time (Do and Faff 2010), our findings indicate that jurisdictional "venue noise" continues to provide fertile ground for arbitrage.

To examine the risk exposure of the portfolio, strategy returns were regressed against common equity factors. Our results show that the systematic risk exposure is largely hedged, with the strategy maintaining a near-zero Beta throughout the sample period. This confirms that the returns are generated by the closing of the pricing gap (the ratio) rather than directional movements in the broader market. This market-neutral characteristic makes the strategy particularly advantageous over traditional directional investments during periods of high economic uncertainty.

Finally, we address the critical role of market frictions. Previous literature has shown that incorporating direct transaction costs can eliminate much of the profitability in quantitative strategies (Lesmond, Schill, and Zhou 2004; Stoll and Whaley 1983). Our results indicate that while the TQQQ/LQQ3 pair generates robust gross returns, the net performance is extremely sensitive to the transaction cost and bid-ask spread applied per trade. We find that the strategy’s return can quickly be diminished by these impacts which highlights the need for trading platforms that charge low fees.

The remainder of the paper is structured as follows: Section 2 details the Z-score and EMA methodology utilized in the study. Section 3 describes the data. Section 4 outlines the summary statistics and trading statistics. Section 5 reports the performance characteristics, Greeks, and the impact of transaction costs, and finally, Section 6 offers concluding remarks.

2. Methodology

The structural framework of this study is designed to identify and exploit transient pricing inefficiencies between the US and UK equity markets. This section details the mathematical normalization of dual-listed assets, the statistical signalling process, and the rigorous performance metrics used to evaluate the strategy's viability.

2.1 Data Normalization and Ratio Construction

To facilitate a direct comparison between TQQQ (denominated in USD) and LQQ3 (denominated in GBP), we must eliminate currency-driven price variance. we normalize the UK-listed LQQ3 into US Dollars using the contemporaneous GBP/USD exchange rate.

We define the **Adjusted Ratio** (S_t) at time t as:

$$R_t = P_{(TQQQ,t)} / \left(\left(P_{(LQQ3,t)} / 100 \right) * FX_t \right)$$

Where:

- $P_(TQQQ,\ t)$ is the price of the US-listed ETF.
- $P_(LQQ3,\ t)$ is the price of the UK-listed ETF, adjusted by a 100 to account for pence-to-pound pricing conventions.
- FX_t is the GBP/USD exchange rate.

By fixing the observation window to 16:00 UTC, we ensure that both the London Stock Exchange and the Nasdaq are in active session, providing the highest possible liquidity and the most accurate reflection of the "Fair Value" of the underlying index.

2.2 Statistical Modelling: EMA and Dynamic Volatility

The strategy assumes that while the ratio R_t may drift due to jurisdictional liquidity imbalances, it remains tethered to a long-term equilibrium.We model this equilibrium using an **Exponential Moving Average (EMA)**, which acts as a dynamic mean that adapts to shifting market regimes:

$$EMA_t = \alpha R_t + (1 - \alpha) EMA_{(t-1)}$$

The smoothing factor α is derived from a lookback period N, defined as:

$$\alpha = 2/(N + 1)$$

To quantify the magnitude of a divergence, we calculate the rolling **Standard Deviation** σ_t of the ratio over the same period N. This allows the model to expand or contract its "confidence intervals" based on prevailing market volatility.

2.3 Signal Generation and Parameterization

The strategy utilizes a **Z-score** to normalize the distance between the current ratio and its moving average. The Z-score is calculated as:

$$Z_t = \left(R_t - EMA_t \right) / \sigma_t$$

The trading logic is governed by a set of input parameters that act as the strategy's sensitivity variables. In this paper, we treat these thresholds as configurable inputs to facilitate a robustness analysis:

- **Entry Threshold ($k_{(entry)}$)**: A position is opened when the absolute value of the Z-score exceeds this value ($|Z_t| > k_{(entry)}$) . If $Z_t > k_{(entry)}$ then S_t = -1 and if $-Z_t > k_{(entry)}$ then S_t = 1 where S_t is the signal at time t.
- **Exit Threshold ($k_{(exit)}$):** A position is held until the ratio reverts toward the mean, closing when the absolute Z-score falls below this value ($|Z_t| < k_{(exit)}$). When this happens S_t = 0.
- **Transaction Cost (C):** A fixed friction percentage applied to the total capital every time a signal changes, accounting for spreads and commissions.

By treating these values as variable inputs, we can test the strategy's sensitivity to execution quality and statistical significance, ensuring that the observed alpha is not a result of "over-fitting" a specific narrow window.

2.4 Portfolio Construction and Daily Return Attribution

To calculate returns, this strategy first calculates the money made on the ETFs before subtracting a transaction fee.

A. Signal Application

As in our strategy we aim to invest the same amount of capital into each ETF of our strategy, we can calculate our gross daily return by looking at the difference in their price movements before multiplying by our signal to take in account the type of position we chose to take. Since the position held on any day is dependent on the signal generated on the previous day, we take the signal as $S_{(t-1)}$.

The Gross Daily Return ($Re_{(gross,t)}$) is calculated as:

$$Re_{(gross,t)} = \left(\left(P_{(TQQQ,t)}/P_{(TQQQ,t-1)}\right) - \left(P_{(LQQ3USD,t)}/P_{(LQQ3USD,t-1)}\right)\right) * S_{(t-1)}$$

Where:

- $S_{(t-1)}$ is the signal from the previous period (-1, 0, or 1).
- $P_{(TQQQ)}$ and $P_{(LQQ3USD)}$ represent the US and currency-adjusted UK prices, respectively.

This formula captures the divergence or convergence of the ratio. If the signal is 1 (Long US / Short UK), the strategy profits if the US ETF which is currently undervalued does better than the UK ETF and moves the ratio towards the mean. If the signal is -1 (Short US / Long UK), the strategy profits if the UK ETF which is currently undervalued does better than the US ETF and moves the ratio towards the mean. If the signal is 0, no position has been taken and so the strategy makes no losses or gains.

B. Accounting for Transaction Frictions

To transition from gross returns to net daily returns ($Re_{(net,t)}$), we must account for the costs associated with entering or exiting a position. These costs are only applied when the model changes position.

The friction is applied as follows:

$$If\ S_t \neq S_{(t-1)} \Rightarrow r_{(net,t)} = r_{(gross,t)} * (1 - C)$$

$$If\ S_t = S_{(t-1)} \Rightarrow r_{(net,t)} = r_{(gross,t)}$$

This ensures that the model considers the amount of trading happening and shows how the transaction costs and bid/ask spreads that arise from this impact profits.

C. Equity Curve Compounding

The cumulative performance of the strategy is tracked via a cumulative equity (E_t) , which represents the growth of a hypothetical $1 investment over the duration of the study:

$$E_t = E_{(t-1)} * \left(1 + r_{(net,t)}\right)$$

This compounded equity curve serves as the basis for calculating the Maximum Drawdown and the total strategy return presented in the final analysis.

2.5 Performance and Risk Attribution

To evaluate the strategy beyond simple nominal returns, we employ several advanced financial metrics. Each is calculated to assess the risk-adjusted quality of the returns and the degree of market neutrality.

A. Total Return, Annualization

The primary measure of success is the Total Strategy Return ($R_{(total)}$), which represents the percentage growth of the investment from the first day of the sample to the last. This is derived directly from the final value of the cumulative equity curve ($E_{(final)}$):

$$Re_{(total)} = E_{(final)} - 1$$

To make this figure comparable to institutional benchmarks and other asset classes, we convert it into an Annualized Return ($Re_{(ann)}$). Since there are approximately 252 trading days in a year, we use this number when calculating the power to raise $Re_{(total)}$ to.

The formula is expressed as:

$$Re_{(ann)} = \left(1 + Re_{(total)}\right)^{(252/T)} - 1$$

Where T is the actual number of trading days observed in our specific data sample.

B. Daily Volatility and Annualized Risk

To measure the risk profile of the strategy, we first calculate the daily volatility ($\sigma_{(daily)}$). This is defined as the Standard Deviation of the entire series of net daily returns from the start date to the end date of the study.

In order to get the annual volatility ($\sigma_{(ann)}$) we multiply by the square root of the number of trading days in a year:

$$\sigma_{(ann)} = \sigma_{(daily)} * \sqrt{(252)}$$

B. Sharpe Ratio

The Sharpe Ratio measures the excess return per unit of risk, providing insight into the efficiency of the strategy:

$$Sharpe = \left(Re_{(ann)} - Re_f\right)/\sigma_{(ann)}$$

where Re_f represents the risk-free rate.

C. Maximum Drawdown (MDD)

The MDD identifies the strategy's worst "peak-to-trough" decline, representing the maximum cumulative loss an investor would have faced:

$$MDD = min\left(\left(Equity_t - Peak_t\right)/Peak_t\right)$$

D. The Greeks: Beta and Alpha

To verify the Market Neutral claim of the strategy, we calculate the Beta (β) and Alpha (α) against the benchmark (TQQQ):

- **Beta:** Measures the systematic exposure to the broader market:

$$\beta = Cov\left(Re_s, Re_m\right)/Var\left(Re_m\right)$$

Here Re_s is the list of daily strategy returns and Re_m is the list of daily market benchmark returns. A Beta near zero would confirm the strategy is truly isolated from directional market movements.

- **Jensen's Alpha:** Measures the idiosyncratic return generated by the strategy after adjusting for its market risk.

$$\alpha = Re_{(s,ann)} - \left(Re_f + \beta\left(Re_{(m,ann)} - Re_f\right)\right)$$

3. Data

This section details the source of all the data used in this paper, and how it was processed before being used to investigate the pair trading strategy.

3.1 Data Sources

The empirical analysis uses historical market data obtained from Stooq.com, a publicly available financial database providing historical price data for equities, exchange-traded funds (ETFs), currencies and other financial instruments. Three hourly datasets were downloaded covering approximately two years of observations:

- TQQQ, a US-listed 3× leveraged ETF tracking the Nasdaq-100 Index.
- LQQ3, a UK-listed 3× leveraged ETF tracking the Nasdaq-100 Index.
- The GBP/USD exchange rate.

These three datasets provide the information required to compare the pricing behaviour of two leveraged ETFs tracking the same underlying index while accounting for fluctuations in the exchange rate.

3.2 Data Preparation

The original datasets consisted of hourly observations. To ensure a consistent comparison between the two ETFs, only observations recorded at 16:00 UTC were retained. Selecting a single observation each trading day provides a consistent reference point while capturing a period in which both the UK and US markets are simultaneously active for much of the sample.

As TQQQ is denominated in US dollars (USD) while LQQ3 is quoted in pence sterling (GBP), the two price series were standardised before analysis. LQQ3 prices were first converted from pence into pounds sterling by dividing by 100 before being converted into US dollars using the contemporaneous GBP/USD exchange rate.

During data cleansing, a small number of LQQ3 observations were identified as being recorded using an alternative price scale (instead of pence they were recorded in pounds). These observations were adjusted to ensure that all prices were expressed consistently before the currency conversion was applied.

3.3 Data Alignment

Following the time filtering and currency conversion, the three datasets were merged using their common trading date. An inner join was used so that only dates for which observations existed across all three datasets were retained. This ensured that every trading signal was generated from a complete and synchronised set of ETF and foreign exchange data.

The final dataset therefore consists of one observation per trading day containing the prices of TQQQ, the currency-adjusted prices of LQQ3, and the corresponding GBP/USD exchange rate. These variables form the basis for constructing the adjusted price ratio, calculating the Exponential Moving Average (EMA), rolling standard deviation, Z-score trading signals, and all subsequent portfolio performance and risk measures presented in this paper.

4. Initial Results

This section presents the empirical findings of the TQQQ/LQQ3 pairs trading strategy during the sample period. To establish a baseline for performance, we first evaluated the strategy using a fixed set of standard parameters before any advanced optimization was applied.

4.1 Baseline Parameterization

For this initial analysis, the model's sensitivity inputs were configured to represent a moderate, standard mean-reversion setup. The baseline parameters were set as follows:

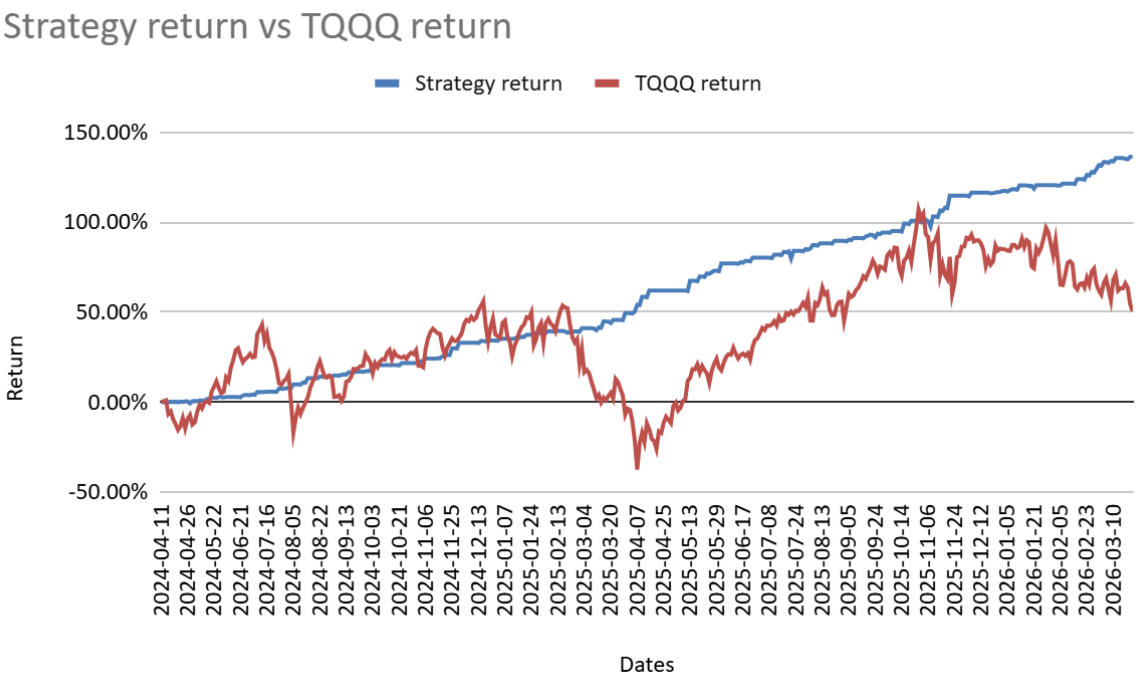
- **EMA Window (N):** 20 periods. This provides a balanced moving average that is responsive to recent price action without being overly sensitive to intraday noise.
- **Entry Z-Score:** 0.75. A position is initiated when the ratio deviates by three-quarters of a standard deviation from the mean, representing a moderate pricing dislocation.

- **Exit Z-Score:** 0.25. The position is held until the ratio reverts to within a quarter of a standard deviation from the mean, ensuring the bulk of the convergence is captured.
- **Transaction Cost:** 0.15% per trade. This friction accounts for the bid-ask spread and institutional commission structures across both the US and UK legs.
- **Risk-Free Rate:** 4.00% (Annualized). This reflects a standard baseline yield for short-term government treasuries during the period, used to calculate excess returns.

Using these baseline inputs, the strategy generated the following performance profile.

4.2 Absolute Returns: Total and Annualized

- **Total Strategy Return:** 136.34% over 405 days for which clean data was available
- **Annualized Return:** 70.77%
- **Annualized TQQQ Market Return (Benchmark):** 29.05% during the period



The return of 136.34% demonstrates that the model successfully identified and exploited pricing inefficiencies between the US and UK markets. Generating a return of this magnitude over the sample period indicates that the strategy did not rely on a few anomalous outlier events, but rather captured a consistent, high-frequency series of spread convergences. Every time the EMA/Z-score threshold was

breached, the model effectively synthesized a profit from the structural misalignment of the two identical indices.

When compounded and scaled to a standard 252-day trading year, the strategy yields an Annualized Return of 70.77%. This 70.77% figure is highly significant when viewed alongside the underlying benchmark. During the same period, simply buying and holding the TQQQ generated an annualized return of 29.05%. The strategy outpaced this long-only, highly directional benchmark by more than double (a spread of over 41 percentage points). This outperformance implies that the strategy's returns are not simply a byproduct of a broader bull market in the tech sector, but rather the result of active, uncorrelated alpha generation.

Crucially, these returns are reported strictly net of the 0.15% transaction cost applied to every directional change in the signal. The fact that the total return remained at 136.34% proves that, with this specific set of parameter inputs, the magnitude of the US/UK pricing gap was wide enough—and the average profit per trade was large enough—to aggressively overcome the persistent drag of trading fees and bid-ask slippage.

4.3 Risk Profile: Volatility and Drawdown

- **Annualized Volatility:** 10.37%
- **Benchmark Volatility (TQQQ):** 69.17%
- **Maximum Drawdown:** -2.21%

To accurately evaluate the strategy's triple-digit returns, one must rigorously examine the risk profile required to achieve them. The primary challenge in trading TQQQ and LQQ3 is their 3x daily reset mechanism, which makes the underlying assets inherently hyper-volatile. Typically, a long position in these leveraged ETFs exposes an investor to extreme price swings; however, the strategy’s Annualized Volatility was remarkably low at just 10.37**%**. This stands in stark contrast to the 69.17% volatility observed in the TQQQ benchmark over the same period.

This divergence is academically significant. It indicates that the long/short construction of the pair successfully neutralized movements of the ETFs they were based on. By simultaneously holding long and short positions across the two jurisdictions, the strategy effectively "cancels out" the systemic market moves, leaving the investor exposed only to the relative pricing gap (the idiosyncratic risk). The transition from nearly 70% volatility down to approximately 10% volatility demonstrates the efficiency of the statistical arbitrage in consistently achieving similar returns.

This risk reduction is further validated by the Maximum Drawdown of just -2.21%. In the context of leveraged instruments, a peak-to-trough decline of merely 2.21% is an exceptional result. A drawdown this shallow implies that the strategy rarely suffered consecutive losses, and when the ratio did move against the model’s position, it did not deviate far before mean-reverting.

From a portfolio management perspective, this "pain threshold" of approximately 2% makes the strategy highly scalable. It suggests returns could be consistently made

without risking a catastrophic margin call or capital depletion. This level of capital preservation is rarely seen in directional equity strategies and highlights the primary benefit of the pairs-trading approach.

4.4 Risk-Adjusted Performance: The Sharpe Ratio

- **Sharpe Ratio:** 6.44

A Sharpe Ratio of 6.44 is an extraordinary outcome and places this TQQQ/LQQ3 pair in a rare category of efficiency. Mathematically, this figure indicates that for every 1% of annualized volatility (risk) the strategy endured, it generated a remarkable 6.44% of excess return above the 4% risk-free rate baseline. This high ratio is a direct consequence of the strategy's low volatility which greatly increases its Sharpe Ratio

Beyond the raw numbers, a ratio this high suggests that the strategy's equity curve is exceptionally smooth, characterized by a steady, linear ascent with very little variance. In statistical terms, it implies a high degree of confidence that the returns are statistically significant and not the result of random chance. It strongly supports the hypothesis that the strategy is capturing a persistent, structural pricing dislocation between the US and UK jurisdictions resulting in a risk-adjusted profile that far exceeds traditional long-only or even most sophisticated long/short equity portfolios.

4.5 Market Neutrality: Correlation and Beta

- **Correlation to TQQQ:** -0.0413
- **Beta vs. TQQQ:** -0.0062

The primary objective of a pairs trade is to function as a pure "Equity Market Neutral" (EMN) strategy. The efficacy of this neutralization is quantified through Correlation and Beta, both of which indicate a near-total decoupling from the broader market. The Correlation coefficient of -0.0413 demonstrates that there is practically zero linear relationship between the daily price movements of the Nasdaq-100 (as represented by TQQQ) and the returns of the strategy. While the market may be trending strongly in either direction, the strategy's equity curve remains independent, moving according to the internal mechanics of the spread rather than the external pressure of the index.

Furthermore, a Beta of -0.0062 suggests that the strategy carries no systematic market risk. Since -0.0062 is statistically indistinguishable from zero, if the broader technology sector were to crash or rally significantly in a single session, the mathematical expectation for the strategy's movement—based on market exposure alone—would be virtually non-existent.

This lack of sensitivity implies that the profit is derived purely from the convergence of the jurisdictional pricing gap, which protects the portfolio against broader macroeconomic shocks, interest rate pivots, or systemic equity sell-offs. This provides a diversification benefit as it lets it generate aggressive returns during periods of market growth or stagnation.

4.6 Idiosyncratic Edge: Jensen's Alpha

- **Alpha (Annualized):** 66.93%

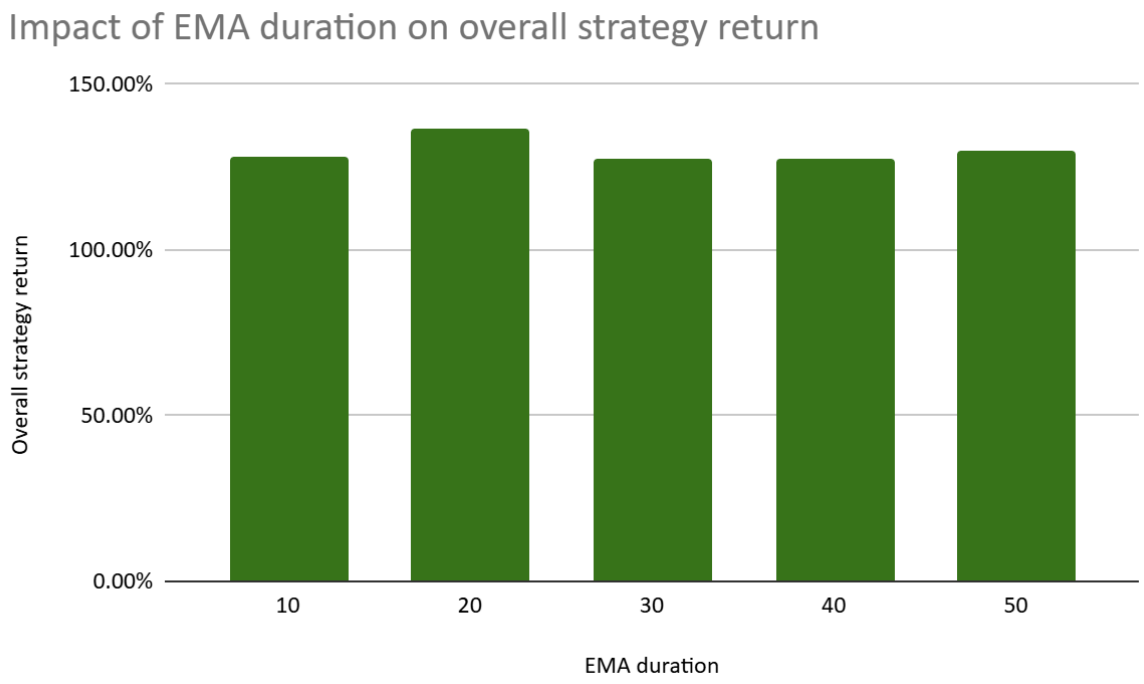
Jensen's Alpha serves as the definitive statistical proof of the strategy's idiosyncratic edge, providing a precise measure of the value added by the model beyond what could be earned through simple market exposure. By isolating the return generated specifically by the pairs trading logic, alpha removes the noise of broader market movements and the baseline risk-free rate. An annualized alpha of 66.93% provides a rigorous confirmation that the moving average and Z-score parameters are capturing significant, repeatable pricing inefficiencies that far exceed the performance of the general market. This helps to quantify the true magnitude of the opportunity available when pairing two leveraged ETFs across the UK and the US.

5. Robustness Check and Sensitivity Analysis

A critical step in validating any algorithmic strategy is the robustness check. The goal is to determine if the strategy's performance is a result of a specific, narrow set of parameters—commonly known as over-fitting—or if the underlying edge is persistent across various configurations. By varying the EMA duration, the Z-score spread, and the transaction costs, we can identify the strategy's breaking point and its optimal operating range.

5.1 Sensitivity to EMA Lookback Duration

The Exponential Moving Average (EMA) serves as the strategy's anchor, defining the "fair value" from which the ratio deviates. We tested the strategy's total return across five different lookback periods to observe how the speed of the mean-reversion signal impacts profitability.



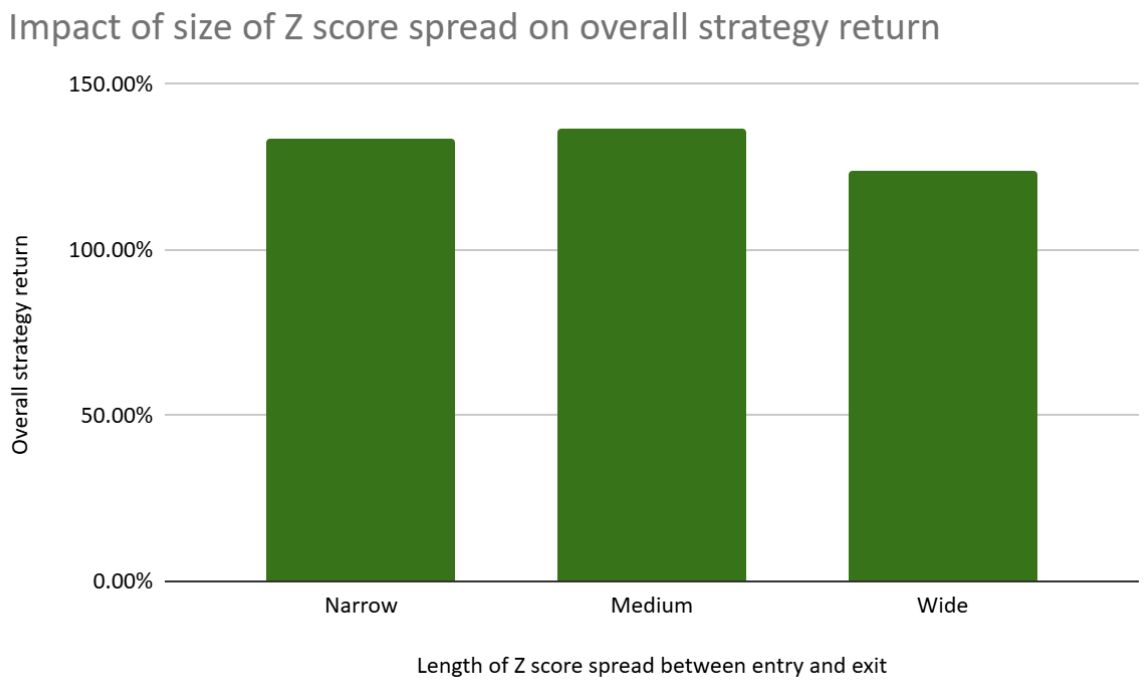
I observed that with different EMAs we received these returns:

- EMA 10: 128.19%
- EMA 20: 136.34%
- EMA 30: 127.59%
- EMA 40: 127.58%
- EMA 50: 129.76%

The results indicate that the strategy is remarkably stable across a wide range of lookback periods, with returns consistently hovering between 127% and 136% which helps us alleviate fears that the strategy had been overfitting. At an EMA of 10, the model is perhaps too reactive to short-term noise, leading to premature entries. Conversely, as the EMA extends toward 50, the mean becomes slower to react to structural shifts in the ratio, which slightly dampens the return. However, the fact that even a 50-period EMA yields nearly 130% suggests that the jurisdictional spread is a persistent phenomenon that does not require perfect timing to exploit.

5.2 Sensitivity to Z-Score Thresholds (Spread Width)

The Z-score thresholds determine the "trigger" for entering and exiting trades. A narrow spread (0.6 entry / 0.4 exit) seeks to capture smaller, more frequent fluctuations, while a wide spread (0.9 entry / 0.1 exit) waits for more significant dislocations.



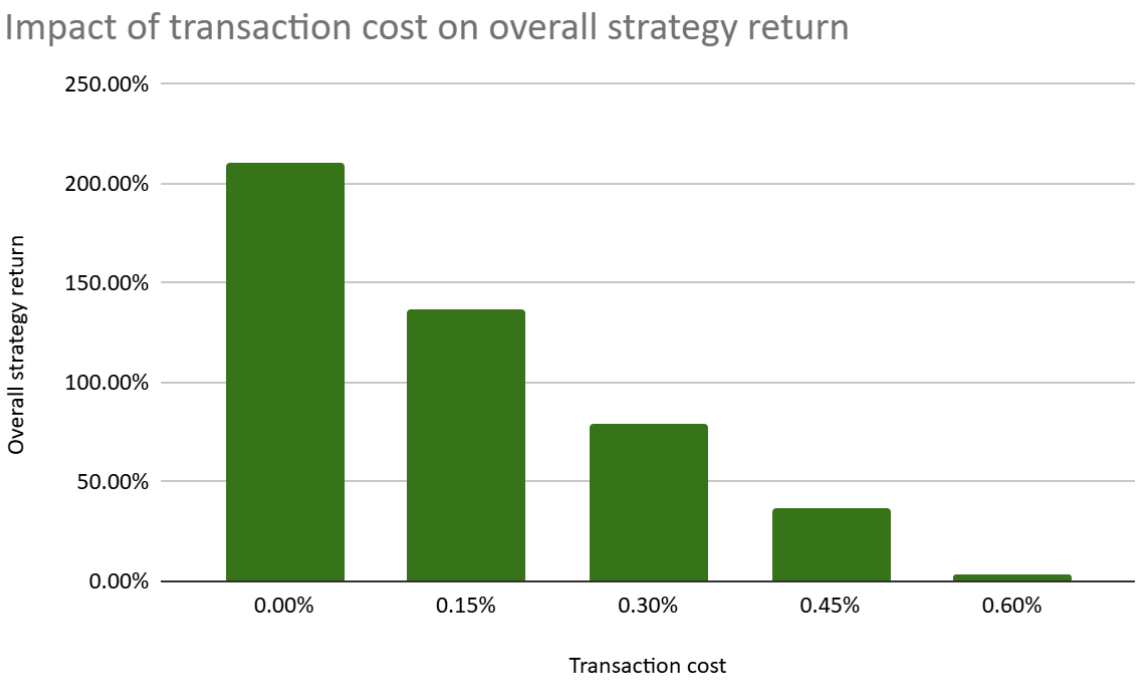
I observed that with different EMAs we got these returns:

- Narrow (Entry 0.6, Exit 0.4): 133.35%
- Medium (Entry 0.75, Exit 0.25): 136.34%
- Wide (Entry 0.9, Exit 0.1): 123.71%

The medium spread of 0.75/0.25 represents the optimal balance for this strategy. The narrow configuration (133.35%) performs admirably but likely suffers from increased transaction costs due to a higher frequency of trades. By tightening the exit to 0.4, the strategy leaves potential profit on the table by closing positions before the ratio has fully reverted to the mean. On the other end of the spectrum, the wide configuration (123.71%) suffers from opportunity cost. By waiting for a 0.9 Z-score to enter, the model misses several smaller but profitable mean-reversion events. Furthermore, an exit at 0.1 requires a near-perfect return to the mean, which may keep the strategy in a trade for too long, exposing it to unnecessary risk. The results suggest that while the best strategy for jurisdictional arbitrage lies in capturing moderate deviations that occur frequently enough to compound capital rapidly, significant profits can be earned even when spreads are narrower or wider

5.3 Sensitivity to Transaction Costs (The Friction Limit)

Transaction costs are often very harmful for high-frequency pairs trading. We tested the strategy against five levels of cost to find how the strategy reacted to market frictions.



- 0.00% Cost: 211.00%
- 0.15% Cost: 136.34%
- 0.30% Cost: 79.52%
- 0.45% Cost: 36.31%
- 0.75% Cost: 3.46%

This analysis reveals the strategy's extreme sensitivity to execution quality. In a frictionless environment (0.00% cost), the strategy is hyper-profitable, returning 211%. However, as costs increase, the decay in performance is non-linear. At 0.15%, our baseline, the strategy remains highly viable at 136.34%. By the time costs reach 0.45%, the return drops to 36.31%, which barely outperforms the buy-and-hold TQQQ benchmark while carrying significantly more operational complexity.

At 0.75% cost—which might represent a retail investor paying high commissions and wide bid-ask spreads—the strategy effectively breaks, returning only 3.46%. This shows that the TQQQ/LQQ3 arbitrage is a strategy that depends heavily on the ability to execute trades with low slippage and minimal commissions typical of institutions rather than retail investors. The rapid decay in returns as costs rise highlights that the spread being captured is relatively thin, requiring a high-volume, low-cost execution environment to remain profitable.

5.4 Robustness Conclusion

Overall, the robustness checks confirm that the strategy is statistically sound. It is not overly dependent on a specific moving average window or a hyper-specific Z-score entry. However, the cost of execution is a significant factor that can greatly affect returns. So long as transaction costs remain below 0.30%, the strategy provides a superior risk-adjusted return compared to the benchmark, validating the core thesis of cross-border jurisdictional arbitrage.

6. Concluding Remarks

We have thoroughly analysed the profitability of a pair trading strategy between TQQQ and LQQ3, two ETFs that track the same index but in different countries. We have discovered that by using a mean-reversal strategy we can exploit fluctuations in the price of the ETFs. This can lead to impressive returns which occur both in bull and bear markets. When we analysed how robust the strategy was we found that while factors such as the EMA duration and the Z score spread seemed to hardly affect the strategy's return, the amount of transaction cost was crucial with small changes greatly changing profits. This shows that there is significant potential in cross-country pair trading strategies, and that more research is needed to analyse their potential profitability.

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Appendix A - Github

The dataset, Python implementation, and supplementary materials for this study are hosted in the public repository: https://github.com/aaravshah0608/Pairtrading_Strategy_Research.

The repository includes the Jupyter notebook code used for modelling the strategy and a guide for reproducing the results.